

Advanced exponential inequalities

№1. Solve the following inequalities:

$$1. \quad \left(5 - \sqrt{24}\right)^x + \left(5 + \sqrt{24}\right)^x \geq 98; \quad 2. \quad \left(\sqrt[3]{4 + \sqrt{15}}\right)^x + \left(\sqrt[3]{4 - \sqrt{15}}\right)^x < 8.$$

№2. Solve the following inequalities:

$$1. \quad x^2 \cdot 2_1^x > x^2 + 2^x; \quad 3. \quad 25 \cdot 2^x - 10^x + 5^x > 25.$$

$$2. \quad x^2 \cdot 3^x + 9 < x^2 + 3^{x+2};$$

№3. Solve the following equations:

$$1. \quad |3^x - 1| + |3^x - 9| = 8; \quad 2. \quad |2^x - 1| + |2^x - 2| = 1.$$

№4. Solve the following inequalities:

$$1. \quad 5^x > 6 - x; \quad 3. \quad 5^x + 12^x < 13^x.$$

$$2. \quad 10^{4-x} > 7 + x;$$

№5. Solve the following inequalities:

$$1. \quad (2^x - 2)\sqrt{x^2 - x - 6} \geq 0; \quad 2. \quad (3^x - 9)\sqrt{x^2 - 2x - 8} \leq 0.$$

№6. Solve the following inequalities for all values of a :

$$1. \quad (x - a)\sqrt{3 \cdot 2^x - 2 \cdot 3^x} \geq 0; \quad 2. \quad (x - a)\sqrt{6 \cdot 5^x - 5 \cdot 6^x} \leq 0.$$

№7. Find all values of a for which the inequality holds for all real x :

$$1. \quad 4^{\cos x} - 2(a - 3) \cdot 2^{\cos x} + a + 3 > 0; \quad 2. \quad (a + 2) \cdot 4^{|x-1|} - 2a \cdot 2^{|x-1|} + 3a + 1 > 0.$$